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The palaeomagnetism of the central zone of the Lewisian foreland, north-west Scotland

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Summary. Three principal directions of magnetization are recognized in the central part of the Lewisian metamorphic terrain of north-west Scotland. The first ('A') magnetization is a high blocking temperature component residing in magnetite and imposed during post-Laxfordian uplift and cooling. Fifty sites yield an overall mean $D = 285.9°$, $I = 54.9°$ and palaeomagnetic pole at 273.2° E, 37.6° N ($dp = 3.7°$, $dm = 5.2°$); this magnetization was probably acquired at crustal depths of 6–10 km and is linked to K–Ar uplift ages averaging 1650–1625 Ma. The second ('B') magnetizations are defined by E–W directions and also reside in high blocking temperature components; they are, however, dipolar, have some properties distinct from the 'A' magnetizations, and are correlated with late stages in the history of the complex at 1400–1200 Ma. The third ('C') NE directed magnetizations reside predominantly in low blocking temperature components in pyrrhotite and possibly maghemite, and were probably acquired at a late stage of the regional uplift; they do not correlate with post-1450 Ma magnetizations from the Laurentian Shield and probably relate to the as yet undefined interval 1600–1450 Ma. The collective palaeomagnetic data and certain geologic data suggest that the Lewisian foreland should be rotated by 30° clockwise about a local axis of rotation on the conventional reconstruction of the North Atlantic continents; this rotation is associated with Lower Palaeozoic transcurrent movements and may be related to a fourth ('D') magnetization of viscous origin.

A collective assessment of 1850–1600 Ma palaeomagnetic data for the Laurentian Shield defines a large apw loop; there is widespread agreement between data from the constituent structural provinces of the Shield although different metamorphic regions define complementary segments of the loop related to uplift over different intervals of time.

1 Introduction

The present decade has seen an extension of the general field of Precambrian palaeomagnetism into the study of metamorphic basement regions. It is recognized that these

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terrains retain a complex record of magnetizations acquired at different times. The cooling of these rocks has generally taken place so slowly that components of secular variation will be averaged and under favourable circumstances apparent polar wander (apw) movement may be recorded within individual samples (Morgan 1976a). While these terrains can, therefore, yield a very valuable record of apw movements, they present at first a major practical problem of evaluating techniques to recover the complete record of magnetizations present and, secondly, an interpretational problem in relating these magnetizations to the complex geochronological record of Precambrian metamorphism, uplift and cooling.

This paper reports a palaeomagnetic study of such a terrain comprising the central part of the Lewisian foreland of NW Scotland. In Proterozoic times this formed a peripheral part of the much larger Laurentian Shield, a cratonic region comprising the Precambrian basement of North America and Greenland (Bridgwater, Watson & Windley 1973).

2 Geology

2.1 REGIONAL SETTING

The Precambrian Lewisian basement of NW Scotland, which includes the mainland north of Skye and the Outer Hebrides, forms a foreland zone to the Caledonian orogenic belt (*c.* 550–380 Ma). It comprises quartzo-feldspathic gneisses affected by amphibolite/granulite metamorphism during an early Scourian episode of crustal incrementation and metamorphism, known as the Badcallian, at approximately 2900–2700 Ma (Moorbath, Welke & Gale 1969); little new material has been added to the complex since this episode (Watson 1975). More restricted reactivation of the Lewisian took place during a late Scourian (Inverian) episode at *c.* 2300–2200 Ma, and widespread intrusion of NW–SE dykes took place in late- and post-Inverian times; these are predominantly tholeiitic in composition but include a later suite of picrite dykes in addition to a number of other compositions. The dykes were first used by Sutton & Watson (1951) to separate a pre-dyke (Scourian) mobile episode from a post-dyke (Laxfordian) mobile episode. They are now wholly or partly amphibolitized, and are foliated to variable degrees in those parts of the Lewisian affected by the Laxfordian episode, although some relict igneous textures are usually present.

The Laxfordian episode included widespread metamorphism predominantly to an amphibolite facies and associated with migmatization and intrusion of granites and pegmatites; it probably reached a climax at about 1850 Ma but a protracted history of uplift and cooling followed the event. In the type area of Loch Laxford the Laxfordian comprises Lewisian rocks affected by episodic thermal/tectonic activity between *c.* 2800 and 1700 Ma and contrasts with the adjacent Scourian terrain where regional thermal/tectonic activity ceased at about 2700 Ma (Watson 1975).

2.2 SCOURIAN–LAXFORDIAN RELATIONSHIPS AND HISTORY OF UPLIFT

The relationship between these two contrasting zones is critical to the interpretation of palaeomagnetic data from the two regions and to linking these data with the history of uplift. The early work of Peach *et al.* (1907) suggested that the parent rocks of the Laxfordian complex were similar to the Scourian complex, a view supported by Sutton & Watson (1951); these workers attributed the distinctive characteristics of the Laxfordian zones to deep seated reworking in post-Badcallian times. Davies & Watson (1977) drew comparisons between a wide range of features in the Scourian complexes and their reworked equivalents in the Laxfordian, but there is no conclusive evidence bearing on the nature of

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vertical movements between the Scourian and Laxfordian, and the geochemical differences between the two areas might result either from a difference in crustal level (Tarney, Skinner & Sheraton 1972; Sheraton, Skinner & Tarney 1973) or to migmatization (Sutton & Watson 1951) associated with granite and pegmatite injection near the climax of the Laxfordian episode.

The metamorphic assemblages within the dyke suite show that they were intruded into a hot crust, probably at very great depths (O'Hara 1961; Tarney 1968; Dearnley 1973). Davies & Watson (1977) quote a minimum depth of 20 km in late Scourian/early Laxfordian times and O'Hara (1975) suggests formation pressures of 13 kbar for Scourian granulites; if this pressure is lithostatic it would be equivalent to crustal depths of 30–40 km (Watson 1977). Tarney (1968) suggests that the dyke emplacement took place at depths of 15–20 km and temperatures of *c.* 500°C. Dickinson & Watson (1976) have also examined disequilibrium metamorphic reactions as an indicator of crustal depth, and identified a trend (which includes the episodes of dyke intrusion) of progressively declining pressures during successively later metamorphic events, with temperatures declining significantly only towards the close of the Laxfordian event; a tentative vertical displacement of 15–20 km is indicated with a cover of thickness of 10–15 km by late Laxfordian times. This latter thickness had clearly been completely eroded by the time of deposition of the Stoer Group (Torridonian) sediments at *c.* 1000 Ma.

2.3 GEOCHRONOLOGY*

The difference in age between the Scourian and Laxfordian episodes was first defined by Giletti, Moorbath & Lambert (1961); these workers dated Scourian and Laxfordian pegmatites at 2490 and 1660 Ma and defined the broad interval during which the dyke intrusion took place. Evans (1964) subsequently dated pegmatites of the Lochinver area at 2335 Ma (U–Pb) thus narrowing these limits, and in a further study dealing specifically with the dykes of the Assynt area (covered by this study) Evans & Tarney (1964) dated a picrite dyke near Lochinver at 2273; K–Ar ages for dolerite dykes predating the picrites ranged from 3860–1550 Ma but the age of intrusion is restricted by the other data to the approximate interval 2335–2273 Ma. Of particular relevance to the palaeomagnetic study are the K–Ar ages of 1630–1390 Ma from completely amphibolitized rocks in areas affected by Laxfordian events. The thermal history of late- and post-Laxfordian times may well have been a complex one; in the Outer Hebrides for example this interval is recorded by Rb–Sr mineral ages of 1654–1483 Ma (Lambert, Myers & Watson 1970) and by K–Ar whole rock ages of 1729–1510 Ma (Lambert, Evans & Dearnley 1970). In the southern part of the Lewisian mainland, Moorbath & Park (1972) date three events following dyke intrusion, the last of these being a pegmatite emplacement at 1420 Ma; indeed mineral age determinations from the Laxfordian show a scatter of dates down to 1400 Ma and locally to 1100 Ma suggesting a complex, and possibly diachronous, history of uplift (Watson 1975).

2.4 SCOPE OF THE PRESENT SURVEY

The present study covers that part of the Lewisian foreland distinguished as the Central Zone (Peach *et al.* 1907; Sutton & Watson 1951) where Scourian structures are extensively preserved; the effects of the Laxfordian episode are largely restricted to narrow shear belts and the dykes are normally discordant with the pre-dyke foliation. The sampling was

* All ages quoted in this paper have been recalculated to the now internationally accepted decay constants given by Steiger & Jäger (1977).

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concentrated on the mafic units, mainly dykes, both because these are likely to have the highest ferrimagnetic content and because they possess the least tectonic fabric; since corrections to the magnetic vector become very considerable for large anisotropy of susceptibility (Beckmann, Olsen & Sorensen 1977) and since the contribution to the resultant susceptibility of the constituents carrying the stable remanence is uncertain (Irving & Park 1972a), it appears advisable to avoid highly anisotropic rocks in magnetic directional studies of metamorphic complexes.

Because the regional temperatures remained high until the close of the Laxfordian thermal/tectonic episodes, there are no grounds for regarding any magnetizations recovered from the Lewisian rocks as relating to the time of intrusion, indeed it is unlikely that any such magnetizations were ever imposed because, under conditions of elevated temperature followed by protracted cooling, the critical range of blocking temperatures is greatly lowered (Pullaiah *et al.* 1975) and will be much lower than the blocking temperature range defined by short-term laboratory experiments. It is probable that regional uplift following the Laxfordian tectonism was so slow that blocking temperature ranges were comparable to the temperatures at which common metamorphic minerals become closed isotopic systems. K–Ar ages for hornblende and biotite in areas of Laxfordian metamorphism average 1650–1625 Ma (Dickinson & Watson 1976); a temperature of 250°C might be assumed for this time but it would be unwise to correlate magnetizations and K–Ar ages precisely both because the actual blocking temperatures are unknown and because the effects of slow cooling on the closure of isotopic systems are only partly understood.

3 Palaeomagnetic results

3.1 THE PALAEOMAGNETIC COLLECTION

The location of the palaeomagnetic sites is given in Fig. 1 and the National Grid references are listed in Table 1. One site typically comprises five to seven samples distributed over several metres of outcrop and drilled in the field; most orientations were made by taking sights on the Sun and the remainder were made using magnetic compass in conjunction with topographic sights.

It is necessary to evaluate the most appropriate practical cleaning techniques for treating slowly cooled rocks for each individual collection. Although these rocks have acquired their magnetic movements during a prolonged thermal process, alternating field (af) cleaning may be equally or more effective than thermal cleaning because the coercivity of magnetic grains is related to their blocking temperatures by the formulae for the relaxation time of a single domain grain (McElhinny 1973). Thermal demagnetization is, however, the only technique which can yield information on the spectrum of blocking temperatures present and is also the only effective method for recovering magnetic components residing in hematite.

The thermomagnetic studies (data summarized in Tables 1 and 2) show that the magnetizations of the Lewisian complex reside predominantly in magnetite or minerals with lower blocking temperatures, although hematite has been identified at a few sites. The af procedure has accordingly been employed here as the principal method for treating this collection although thermal demagnetization of one or more samples from each site has been undertaken to define blocking temperature ranges and to ascertain whether components had been missed by the af method. The af demagnetization was undertaken with a 500 c/s system through which the ambient magnetic field was regularly and carefully annulled, and it utilized steps of 5 to 10 milli-Tesla (mT) up to peak field of 150 mT or to fields at which the magnetizations could no longer be accurately measured; thermal demagnetization was undertaken in steps of 100°C reduced to 30°C near the Curie points using a commercial

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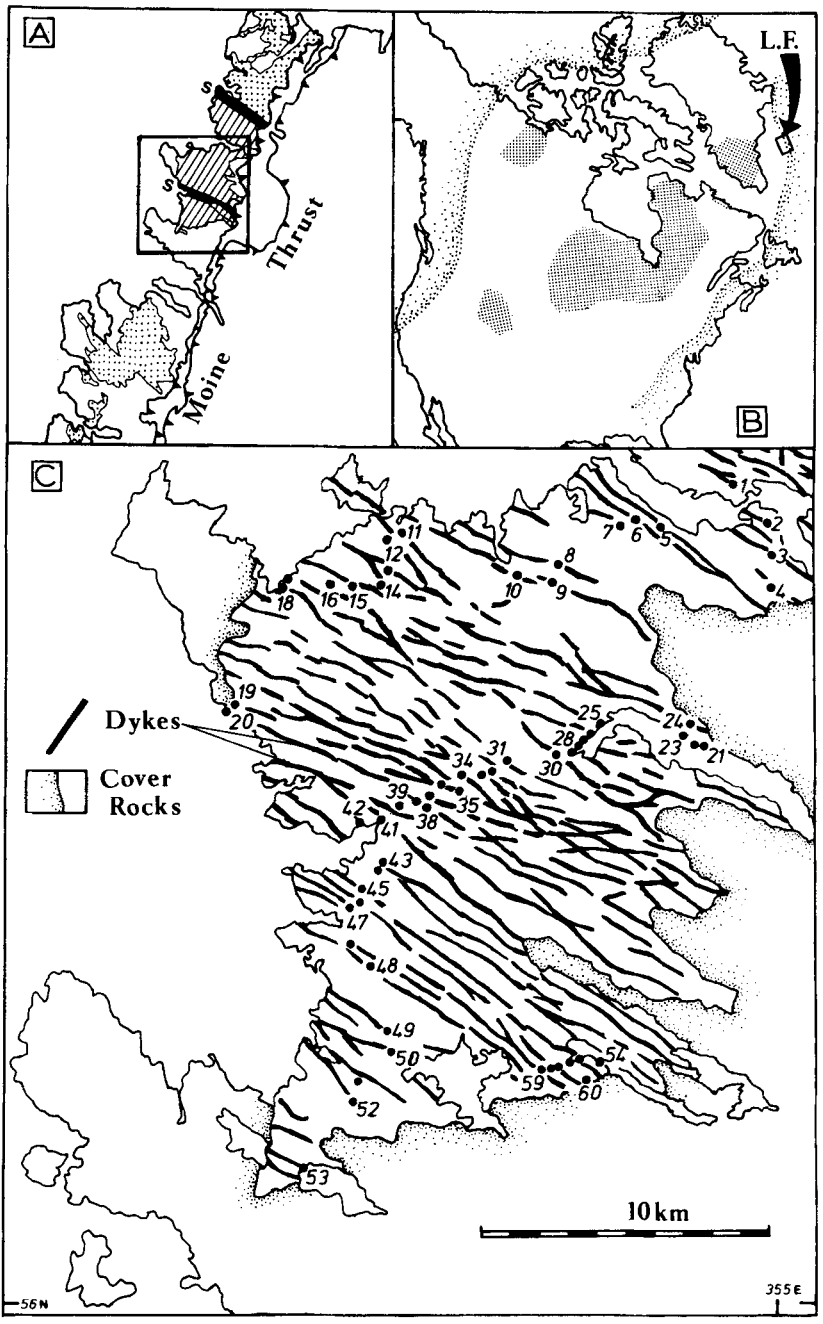


Figure 1. (a) The Lewisian foreland of NW Scotland with the area of study indicated. The Scourian metamorphic assemblages are shaded and the Lewisian assemblages are stippled; S = major shear zone. Later rocks are left blank. (b) Reconstruction of the Laurentian Shield with the position of the Lewisian foreland (LF) indicated; the Archaean nuclei are stippled. (c) Palaeomagnetic sample localities of this study in the central part of the Lewisian outcrop (a); the distribution of basic dykes is generalized after Geological Survey ¼ in sheet No. 5.

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Schonstedt demagnetizer. In addition a number of samples were subjected to a two-stage treatment (Park 1975) of af followed by thermal treatment. Thermomagnetic determinations (saturation magnetization, J_s, versus temperature) were found to be particularly useful in identifying separate components of magnetization; one determination was made from each site and additional determinations were made where more than one component of magnetization was identified.

3.2 PALAEOMAGNETIC RESULTS

The majority of the sites from this collection (40 out of 62) yield either one single direction of magnetization with a moderate positive inclination and westerly declination, or have this direction as one component. The sites in which this is the only component are typically highly stable to af demagnetization and exhibit slight and systematic movement over the range of treatment (0–150 mT); at low fields of 5–25 mT samples cease to move along a great circle path through the present Earth's field direction (Fig. 2) and sample directions have been compounded at this stage to calculate site means (Table 1). These magnetizations isolated by af treatment are stable to subsequent thermal demagnetization (Fig. 2): the directions do not change and residual moments do not fall appreciably until the Curie points are approached at 530–560°C. A number of paired cylinders from the same field core have been thermally and af cleaned to compare the efficiency of the two techniques in isolating this remanence. Typical examples are illustrated in Fig. 2 (sites 39 and 53): stable end points reached by af demagnetization at 10 mT are not achieved by thermal demagnetization until temperatures of 400–500°C. Morgan (1976b) observed this behaviour in the demagnetization of magnetite-bearing basement rocks, and has utilized the two sets of curves of Evans & McElhinny (1969) relating blocking temperatures and unblocking fields to magnetite grain dimensions to provide a physical explanation: the unblocking field curves

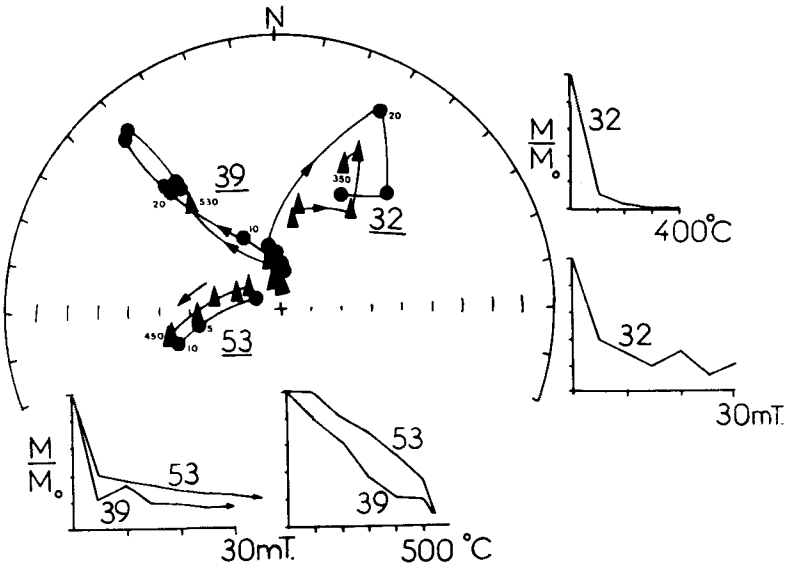


Figure 2. Behaviour of some typical samples exhibiting 'A' and 'C' magnetizations with progressive af (dots) and thermal (triangles) treatment. Projections are stereographic and lower hemisphere (positive inclination) plots are indicated by solid symbols and upper hemisphere (negative inclination) plots are indicated by open symbols. Fields in mT or °C.

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cut across the unblocking temperature curves so that a given af demagnetization field can demagnetize grains with a very wide spectrum of blocking temperatures; in addition, depending on the distribution of grain sizes present, a given af field will selectively demagnetize more grains with lower blocking temperatures than with higher blocking temperatures.

The stable samples with single component magnetizations all give simple thermomagnetic spectra of the type shown in Fig. 4 (sites 24, 62) with single Curie points with the range 515–550°C attributable to titanomagnetite; in most cases, as in the examples illustrated in Fig. 4, the curves are completely reversible (Table 1).

The sites falling in this group with single, high coercivity, high blocking temperature remanence are listed in Table 1 and are given the collective designation of 'A' magnetizations to distinguish them from other less well-defined, and probably younger, magnetizations described below. Twenty-four sites falling in this group (2, 3, 9, 12, 14, 24–26, 28, 30, 34, 38, 40, 45, 49, 50, 52–55, 57, 60–62) yield simple reversible thermomagnetic spectra with a single high Curie point attributable to titanomagnetites. Six further sites in this group (10, 11, 16, 29, 41 and 42) show a significant drop in saturation magnetization below about 300°C (as illustrated by sites 29 and 42, Fig. 4) followed by a single high Curie point curve typical of the other sites; this component must undergo alteration on heating because saturation remanence increases to variable degrees on cooling and the curves are not reversible; no low blocking temperature component has, however, been identified at these sites. By contrast a further 10 sites (5, 8, 13, 19, 20, 22, 37, 39, 44 and 47), all of which have these more complex thermomagnetic spectra, do show a low blocking temperature component in some or all samples from the same site; in most cases this can be removed by progressive af and/or thermal cleaning to recover a group 'A' direction. The significance of these components is considered below but since the 'A' direction is recovered in all cases at the higher field or temperature, it is regarded as representing the primary component and these sites are therefore combined with the remaining 31 sites in Table 1.

The overall mean is $D = 286.0°$, $I = 54.7°$ yielding a palaeomagnetic pole at 272.9° E, 37.5° N ($dp = 4.4°$, $dm = 6.2°$). In an earlier study of the Lewisian dykes, Beckmann (1976) found 10 sites in this area yielding group 'A' magnetizations including all the sites with highest coercivities of magnetization; it is possible that this magnetization was also present in some of his 28 remaining sites, but the demagnetizing apparatus used was inadequate to resolve this in samples with low magnetic coercivities. Although some of the rock units sampled in this study are the same as those sampled by Beckmann it is legitimate to combine them by giving unit weight to each site because the magnetizations derive from metamorphic rocks cooled over long periods of time. The mean (Table 1) of all 50 sites is then $D = 285.9°$, $I = 54.9°$, yielding a palaeomagnetic pole for the central part of the Lewisian foreland of 273.2° E, 37.6° N ($dp = 3.7°$, $dm = 5.2°$).

A second series of sites listed in Table 2 is defined as group 'B', although since the sites belonging to this group are not collectively well grouped their nature is much more difficult to decipher. They are regarded as a high blocking temperature group because they yield directions stable to thermal treatment and high Curie points attributable to titanomagnetite and hematite (sites 17 and 31) but their behaviour to af demagnetization is in general quite different from the group 'A' demagnetization. Instead of moving to a single stable end point sample directions migrate progressively along irregular paths over the whole range of af treatment (Fig. 5). Also two polarities appear to be present: sites 27, 47 and 48 are antiparallel to sites 17, 18 and 19 and sites 4 and 58 are approximately antiparallel to site 17. At sites 19 and 47 group 'A' magnetizations (site 17) are present in separate samples from

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Table 1. Palaeomagnetic results and thermomagnetic properties for sites yielding group 'A' magnetizations. Lewisian metamorphic rocks, NW Scotland; af and thermally cleaned.

Site No.	Nat. Grid Ref.	N	D	I	k	Mo $x_2 10^{-5}$ A.m^2/kg	Thermomagnetic Results
2	22329331	5	303.9	71.3	28	2.96 - 0.03	530, A
3	22359320	6	276.1	47.6	21	0.18 - 0.03	540, A
5	21999300	5	281.7	46.7	77	1.51 - 0.02	545, B, LTC
+8	21619318	3	290.8	43.6	39	0.02 - 0.01	540, B, LTC
9	21579314	6	324.8	68.6	208	30.28 - 1.29	535, A
10	21469315	4	294.9	55.8	55	0.93 - 0.01	520, D, LTC
11	21039328	6	283.5	54.5	14	1.66 - 0.01	510, B, LTC
12	21029317	8	313.7	31.1	69	4.98 - 0.44	560, A
+13	20979308	5	261.5	42.2	23	0.01 - 0.00	540, A, LTC
14	20959317	7	287.9	67.7	23	0.01 - 0.00	560, A
+16	20789315	4	312.5	51.0	9	0.03 - 0.01	530, A, LTC
+19	20459267	4	329.7	59.0	56	0.98 - 0.01	505, A, LTC
20	20449265	7	310.3	54.3	151	0.02 - 0.01	510, B, LTC
+22	22099254	4	255.5	51.6	30	3.64 - 0.27	530, A, LTC
24	22019261	9	311.2	41.0	100	16.50 - 1.76	520, A
25	21739262	7	268.2	48.9	129	28.80 - 9.10	530, A
26	21729260	7	284.6	53.8	66	14.70 - 2.96	525, A
28	21679255	6	292.7	60.9	40	0.49 - 0.01	550, A
29	21639254	8	298.5	59.8	172	3.42 - 0.04	540, D, LTC
30	21559251	7	303.3	55.4	93	5.46 - 3.79	520, A
34	21269247	7	306.5	51.1	39	0.02 - 0.01	530, A
+37	21159241	5	281.2	46.8	16	0.02 - 0.01	555, D, LTC
38	21199239	6	295.3	69.4	29	0.02 - 0.01	540, A
+39	21009239	5	285.6	45.5	10	0.02 - 0.01	540, D, LTC
40	21049307	6	278.6	51.1	25	0.22 - 0.00	540, A
41	20969233	7	308.2	36.3	95	0.01 - 0.00	535, D, LTC
42	20859300	8	322.4	40.2	25	0.11 - 0.01	530, D, LTC
+44	20949215	7	280.2	58.6	12	0.24 - 0.01	520, D, LTC
45	20839212	6	279.1	39.8	10	3.48 - 0.01	520, A
+47	20809205	3	281.2	48.3	63	2.66 - 0.01	530, A, LTC
49	20829185	7	272.4	56.7	287	19.55 - 12.70	545, A
50	20899182	11	259.6	48.6	38	42.59 - 3.68	535, A
52	20959149	6	273.8	53.2	245	12.11 - 2.89	535, A
53	20869142	8	257.5	48.5	197	13.29 - 4.63	515, A
54	20819136	8	275.5	54.5	739	16.20 - 3.27	530, A
55	20629108	6	282.1	66.8	889	27.70 - 23.20	520, A
57	21659145	7	253.2	54.3	168	13.38 - 0.27	540, A
60	21569142	7	239.7	56.5	222	6.10 - 4.60	545, A
61	21559140	7	274.2	58.9	1361	4.48 - 4.47	530, A
62	21619134	6	254.9	61.7	477	7.89 - 0.47	535, A

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Table 1 – *continued*

Group mean, 'A' magnetizations					Palaeomagnetic pole	
N	*D*	*I*	*k*	α_{95}	° E	° N
40	286.0	54.7	27	4.4°	272.9° (*dp* = 4.4°,	37.5° *dm* = 6.2°)
Group mean, including sites of Beckmann (1976)					Palaeomagnetic pole	
N	*D*	*I*	*k*	α_{95}	° E	° N
50	285.9	54.9	31	3.7°	273.2° (*dp* = 3.7°,	37.6° *dm* = 5.2°)

(see footnotes to Table 2)

the same site and both have comparable properties. There is also some suggestion of an overlap between the two groups because a few sites included in group 'A' have a wide range of magnetic inclinations present at one site; this applies notably at site 24 (Table 1) where a narrow range of declinations but a wide range of inclinations result in a low overall value of the precision parameter *k*. Site 48 (Table 2) has shallow negative inclinations and with progressive demagnetization some samples move to steeper negative inclinations; this site also exhibits two distinct Curie points attributable to titanomagnetite (500 and 530°C). The presence of antiparallel directions is noteworthy: Beckmann (1976) considered that the Lewisian rocks were unlikely to have been magnetized during slow cooling because, in a reversing field, they would successively acquire a sequence of antiparallel components as they cooled through their blocking temperature ranges. It is necessary to postulate either an alternative mechanism (Beckmann considered a shock induced remanence related to seismic events) or that the prevailing magnetic field was predominantly of one polarity. Morgan (1976c) considered that a protracted magnetization during slow uplift and cooling is the only tenable mechanism applicable to the Lewisian terrain, and it is the only one which can clearly be related to the geological history outlined in Section 2.

In Table 2 the 'B' magnetizations are combined together into two groups as antiparallel sites 4, 17 and 58 (B1) and antiparallel sites 18, 19, 27, 31, 47 and 48 (B2); it is not suggested that this is the only way of treating the data because they do not form a compact group (Fig. 7) and they may not represent a coherent geological event; to define their real

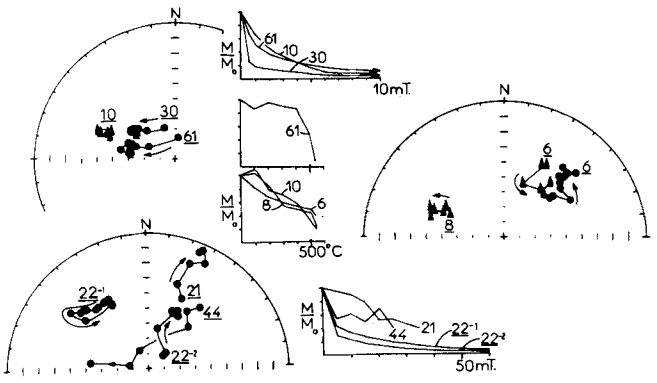


Figure 3. Progressive demagnetization behaviour with af and thermal cleaning, 'A' and 'C' magnetizations. Conventions are the same as for Fig. 2.

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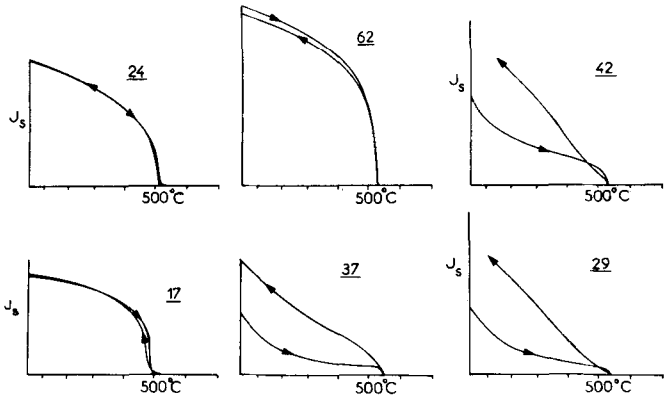


Figure 4. Typical thermomagnetic determinations (saturation magnetization, J_s, versus temperature in °C); see Tables 1 and 2 and text.

significance would require a more detailed sample even than that achieved in the present study. Beckmann (1976) identified a single dyke at Badcall near Scourie with a stable magnetization removed from the group 'A' directions of $D = 106.1°$, $I = 1.0°$ ($k = 35$) and similar to the group 'B' directions of this study; this dyke was dated by Evans & Tarney (1964) at 1873 Ma in its least altered part and 1450 Ma in an amphibolitized region defining probable limits to the age of magnetization.

We may observe in passing the identification of similar 'B' magnetizations at adjacent basic bodies (18, 19, 47, 48 A) and the location of 'B' magnetizations of dyke site 58 between four dykes showing group 'A' magnetizations; this suggests that tracts of country including more than one dyke may record magnetizations of different age from the surrounding terrain, and presumably the temperature at which a magnetization was acquired will depend both on compositional variations and on cooling conditions, in particular the presence of fluids and the proximity to shear zones. However, one further possibility is that a much later phase of dyke intrusion is present within the foreland to account for the 'B' magnetizations; to date no clear petrologic or radiometric evidence for a younger dyke swarm is available.

One further group of directions defined by this study has NE declinations and positive inclinations (Fig. 7) and is classed here as group 'C' with properties summarized in Table 2. Two of the sites falling in this grouping (6 and 59, Fig. 2) have single high blocking temperature components, while at sites 7, 15, 21, 23, 32, 35, 36, 43 and 51 a similar direction is present as the only magnetization present at these weakly magnetized sites (Fig. 3, Table 2). At five sites (8, 20, 22, 39, 44) this component is identified in some samples although 'A' magnetizations are also present at these sites; in general these magnetizations are not identified in the same specimen although a few samples (e.g. site 44, Fig. 2) show a movement to group 'A' directions with progressive demagnetization. Thermal demagnetization and thermomagnetic results show that sites 6 and 59 and probably 20 and 32 have this component residing in titanomagnetite, but this does not appear to be the case at most of the remaining sites which show a rapid and exponential loss of J_s and magnetic moments which become too weak to measure after thermal treatment at 300–400°C (32, Fig. 2).

The low blocking temperature 'C' components might reside in either pyrrhotite ($Fe_{1-x}S$) of maghemite (γFe_2O_3). Since the latter mineral converts continuously to αFe_2O_3 on heating over the range 300–700°C it is a possible contender at least at site 39 (Fig. 6) where a small component of J_s attributable to hematite is observed. Comparable low

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blocking temperature components have also been identified from other Precambrian terrains (e.g. McGlynn & Irving 1978) but they present a particular physical problem because they lie at the lower end of the range of magnetizations which could possibly survive from Lower Proterozoic times. Studies of the variation in relaxation time with temperature (Chamalaun 1964; McElhinny 1973) show that magnetic grains with blocking temperatures of 280°C would, at surface temperatures, have a relaxation time of only 100 Ma while grains with slightly higher blocking temperature of 310°C would have a relaxation time of over 1000 Ma. Although pyrrhotite loses the bulk of its remanence by 300°C (Nagata 1961) it undergoes a Néel-type transition at approximately 310°C (Bhimasankaram 1964) and retains weak ferrimagnetic properties to 560°C. It may thus be considered in the context of Precambrian magnetizations and is a prominent constituent of some of the Lewisian rocks (next section).

In view of the differing origin of the magnetizations falling in the 'C' group, they may be of different ages and it is not strictly correct to combine them into a single mean; also they show a non-Fisherian distribution which is smeared towards the primitive (Fig. 7). However, for the purposes of discussion the 14 better grouped directions with moderate positive inclinations have been combined in Table 2 to yield a mean $D = 42°, I = 40°$ representative of this direction of magnetization.

Four sites (13, 16, 23 and 37) and a few individual specimens from several other sites show a SE positive component of viscous remanent magnetization which is always removed by af demagnetization in low fields (Fig. 5). This direction (mean $D = 161°$, $I = 48°$ (four sites, $k = 115$) palaeomagnetic pole at 11° E, 1° S) is worthy of note because it is the only one from this study which closely correlates with (post-Cambrian) Lower Palaeozoic palaeofield directions for Britain. It is similar to Ordovician palaeofield directions for Britain south of Aberdeenshire and (with the possible exception of sites 46 and 56, Table 2) the only evidence for remagnetization during Palaeozoic orogenic activity within the adjacent Caledonian belt.

4 Petrographic observations

In his earlier study of Lewisian dykes, Beckmann (1976) sought to clarify opaque petrological differences between the rocks carrying low coercivity (classified as 'unstable')

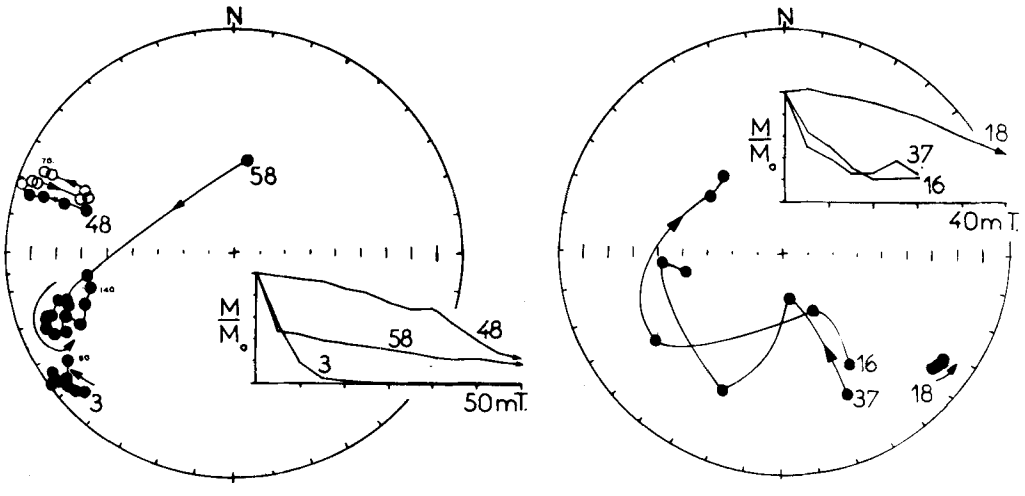


Figure 5. Demagnetization behaviour with progressive af treatment of samples with 'B' and 'D' magnetizations. Conventions are the same as for Fig. 2.

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Table 2. Palaeomagnetic results and thermomagnetic properties of sites yielding magnetizations other than Group 'A' directions. Lewisian metamorphic rocks, NW Scotland.

Site No.	Nat. Grid Ref.	N	D	I	k	Mo x 10^{-5} $A.m^2/kg$	Thermomagnetic Results
GROUP 'B' MAGNETISATIONS							
4	22359310	8	246.1	13.7	18	37.15 - 0.41	530, B
17	20649315	6	85.5	-7.9	45	0.07 - 0.01	495,650, A
18	20639314	7	117.2	15.4	43	1.55 - 0.00	500, A
19	I	5	106.8	30.5	42	I	(505, A)
27	21709258	6	293.4	-22.9	39	2.07 - 0.18	530, A
31	21399250	7	116.7	52.1	12	0.02 - 0.01	505,660, D, LTC
47	I	5	287.9	-35.0	13	I	530, B, LTC
48A	20809205	5	304.9	-18.5	48	0.34 - 0.09	500, 530, A
48B	20809205	5	314.2	-54.3	45	——	500,530, A
58	21649144	6	244.4	4.0	97	21.33 - 1.88	540, A
GROUP 'C' MAGNETISATIONS							
6	21889302	7	52.2	59.1	71	6.60 - 0.25	565, A
7	21829331	4	43.3	26.4	19	0.02 - 0.01	——, A, LTC
8	I	7	52.0	29.4	17	I	540, B, LTC
15	20879314	5	46.0	38.7	18	0.02 - 0.01	——, B, LTC
20	I	4	42.9	48.?	13	I	510, B, LTC
*21	22109252	5	19.3	0.7	39	0.03 - 0.01	——, B, LTC
22	I	4	23.2	21.1	8	I	530, A, LTC
23	22029258	5	48.4	40.1	14	0.02 - 0.01	——, B, LTC
32	21329248	7	20.6	49.6	23	0.08 - 0.02	540, B, LTC
35	21219242	5	58.1	39.3	20	0.46 - 0.01	——, A, LTC
36	21189242	7	59.4	55.4	16	0.03 - 0.01	——, A, LTC
39	I	6	32.3	46.0	21	I	540, D, LTC
*43	20959216	8	12.2	12.3	22	0.03 - 0.01	——, A, LTC
44	I	5	30.7	54.8	47	I	520, D, LTC
51	20959157	3	52.4	11.4	30	0.07 - 0.02	——, A, LTC
59	21619142	7	32.6	27.8	284	0.03 - 0.01	520,640, A
OTHER MAGNETISATIONS OF UNCERTAIN SIGNIFICANCE:							
1	22229345	5	40.7	-60.8	15	7.45 - 0.01	560, A
46	20809207	6	357.6	-18.2	102	25.40 - 1.87	495, A
56	21729103	7	17.9	-15.5	118	0.03 - 0.01	500, D, LTC

		D	I	K	
Group Mean Directions:					
(i)	Lewisian 'B1' directions:				Palaeomagnetic Pole
	3 sites	251.9°	8.7°	41	282.6° E, 5.7° S
		(R = 2.95, α_{95} = 19.6°)			(dp = 9.9°, dm = 19.7°)

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Table 2 – *continued*

(ii)	Lewisian 'B2' directions:				
	7 sites	296.6°	-32.9°	22	233.1° E, 2.1° S
		(R = 5.78, α_{95} = 14.5°)			(dp = 8.4°, dm = 14.8°)
(iii)	Lewisian 'C' directions:				
	14 sites	42.3°	39.8°	23	116.1°E, 43.3°N
		(R = 13.4, α_{95} = 8.5°)			(dp = 6.1°, dm = 10.2°)

Footnotes

D and *I* are the mean declination and inclination of the magnetization derived from *N* samples with precision parameter $K(=(N-1)/(N-R))$ and 95 per cent radius of confidence, α_{95}, about the mean direction; *dp* and *dm* are the radii of the oval of 95 per cent confidence about the palaeomagnetic pole in the colatitude direction and at right angles to it respectively. + sites which also include a low blocking temperature component. The column 'Thermomagnetic results' includes the Curie point in °C (± 10°) and the form of the curve is categorized as A: heating and cooling curves similar, B: heating curve below cooling curve, C: heating curve above cooling curve, D: large increase in saturation magnetization on cooling. LTC = low blocking temperature component suggested by large fall in saturation remanence between room temperature and about 300°C. Mo is the intensity of NRM prior to af or thermal cleaning.

remanence and high coercivity remanence. He found that the main visible distinction occurred in the ilmenite grains which contained abundant lamellae (with little compositional difference from the host grains) in the samples with high coercivities.

The present study was oriented towards possible petrologic differences between the specimens carrying 'A', 'B' and 'C' magnetizations. The 'A' group sites which are both strongly magnetized and highly stable (e.g. 19, 20, 59–62) contain variable amounts of pyrite and two phases of titanomagnetite as large homogeneous grains with little or no marginal alteration and numerous fine rods (*c.* 1 × 20 μm) oriented along cleavage planes in silicate phases; this latter phase is widely reported as a carrier of stable high coercivity NRM (e.g. Evans, McElhinny & Gifford 1968). Other samples exhibit more complex phases with patchy alteration of titanomagnetite to hematite/rutile aggregates (sites 38 and 57) or a clothwork texture with development of spinel rods (11). The weakly magnetized sites in general exhibit a very sparse opaque content while all samples yielding type D thermomagnetic curves exhibit magnetite associated with thick rims of maghemite. In general the

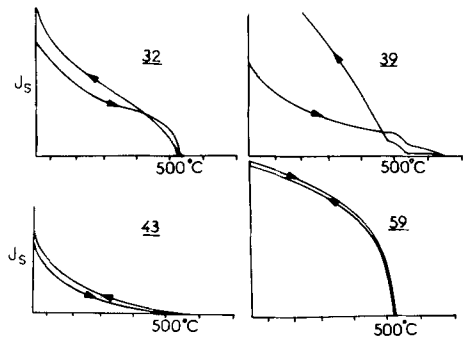


Figure 6. Typical thermomagnetic determinations for sites including group 'C' magnetizations.

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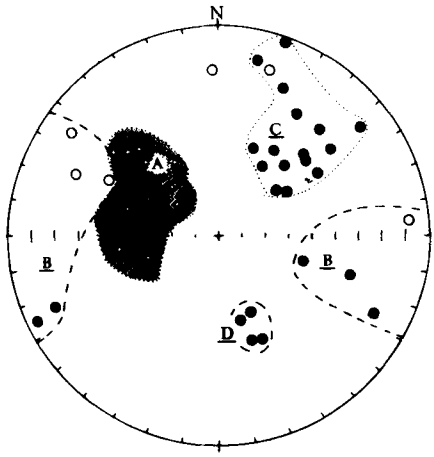


Figure 7. Distribution of 'A', 'B', 'C' and 'D' magnetizations; the field of the 'A' magnetizations is stippled; stereographic projection.

group 'B' sites have similar characteristics to the group 'A' sites, the main distinction being the widespread alteration of magnetite to hematite at sites 17, 31, 47 and 48.

Of the 10 'C' group sites 6 and 59 have comparable characteristics to the highly stable 'A' group samples, but sites 8, 39 and 44 where this magnetization occurs as a second component all have magnetites with large maghemite overgrowths, and it seems likely that the 'C' magnetization resides in this phase. The 'C' group samples which have weak magnetizations and exhibit exponential fall in J_s contain very little opaque material; this is predominantly fine sulphide of which pyrrhotite is an important fraction (e.g. sites 7, 21, 35, 36, 43).

4.1 AGE OF THE MAGNETIZATIONS

The magnetizations recognized in these Lewisian rocks yield the estimates of palaeomagnetic poles listed in Tables 1 and 2; the pole derived from the 'A' magnetization is regarded as the oldest because this magnetization resides in primary titanomagnetite associated with silicate phases. It is not possible to identify with certainty the age relationships of the 'B' and 'C' magnetizations; however, the 'C' magnetizations are of one polarity only and, since they are carried by pyrrhotite at some sites and probably by maghemite at others, they could have been acquired at a late stage in the uplift and cooling of the metamorphic complex or possibly during later reheating.

In the absence of a knowledge of the rate of cooling, it is also not possible to calculate precisely the reduction of the blocking temperature range with slow cooling, but the calculations of Pullaiah *et al.* (1975) show that it is reasonable to assume that the blocking temperature ranges averaging 570–470°C were depressed to *c.* 250–150°C for those rocks cooled during uplift over periods in excess of 10^7yr (Dickinson & Watson 1976). These temperatures would only have been achieved at late stages in the post-Laxfordian uplift and cooling by which time–temperature gradients of 45°C/km achieved at the climax of the Laxfordian episode (Dickinson & Watson 1976) would have been reduced to perhaps 25°C/km. This implies that the magnetizations were imposed at crustal depths of 6–10 km and the higher end of this temperature range (relevant to the 'A' magnetizations) is defined by K–Ar mineral dates averaging 1650–1625 Ma (Dickinson & Watson 1976). The lower end of this

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range (perhaps applicable to the 'C' magnetizations) might not have been reached until 1400 Ma.

4.2 RELATIONSHIP OF LEWISIAN DATA TO PALAEOMAGNETIC RESULTS FROM THE LAURENTIAN SHIELD

To compare the Lewisian magnetizations with contemporaneous results from elsewhere in the Laurentian Craton, the palaeomagnetic poles are listed in Table 3 after rotation according to the continental reconstruction of Bullard, Everett & Smith (1965); this corrects for Phanerozoic drift. Although this reconstruction has been widely used to interpret palaeomagnetic data from NW Scotland, it should not be accepted unequivocally for two reasons: first, although the Ketilidian belt of south Greenland has some analogies with the Laxfordian, no continuation of the Archaean nucleus of central-southern Greenland is identified in the Lewisian foreland; secondly, prior to 390 Ma there is a major discrepancy between the palaeomagnetic results from Britain and from North America/Continental Europe on the Pangaean reconstruction (Morris 1976; Piper, in preparation). This anomaly is only known to apply to Britain south of the Great Glen Fault but it is possible that the hypothesized zones of sinistral strike-slip motion necessary to account for this discrepancy using tectonic models lie outside of the Caledonian belt and within the Precambrian foreland. Indeed, in a major reassessment of Upper Proterozoic palaeomagnetic data Stearn (1978) finds that these Upper Proterozoic palaeomagnetic data only agree precisely with contemporaneous Laurentian and Baltic Shield data if the Lewisian foreland is rotated by 30° clockwise about a local axis of rotation. Such a motion is consistent with the large-scale strike-slip motions along the orthotectonic Caledonides inferred from Palaeozoic data and should be identifiable from the alignment of major Precambrian features within the foreland. The Scourie dykes (average trend 120–125° E) are part of a larger conjugate swarm found in Greenland and Labrador (Escher, Jack & Watterson 1976). In west Greenland these dykes have a trend averaging 100° E while in the same (Nagssugtuqidian) belt in east Greenland the trend is 120–125° E. On the Bullard *et al.* (1965) reconstruction the Scourie dykes strike approximately 20° further to the east of the east Greenland dykes, but if the foreland is rotated by 30° clockwise this is reduced to 10° to the west. There is thus some geological evidence for Stearn's (1978) conclusions from the palaeomagnetic data, and the Lewisian palaeomagnetic pole is plotted in Fig. 8 both before and after a local rotation of 30°. It will be noted that the Lewisian 'A' magnetizations then agree more closely with the contemporaneous Ketilidian poles from south Greenland.

4.3 MIDDLE-LOWER PROTEROZOIC APW RELATIVE TO THE LAURENTIAN SHIELD

To evaluate the significance of the Lewisian results in the context of Laurentian palaeomagnetic poles, the collective data for the interval 1850–1400 Ma are tabulated in Table 3 and plotted in Fig. 8. Since no magnetization from the Lewisian can predate the Laxfordian episode, all data known to be older than 1850 Ma are excluded. However, even this selection is somewhat subjective because palaeomagnetic results from cross-cutting intrusions have been variously linked to Rb–Sr isochrons or Rb–Sr and K–Ar mineral ages, and the evidence as to whether most Lower Proterozoic anorogenic intrusions were emplaced into country rock with regional temperatures above or below the Curie point is often of an equivocal nature. The section of Table 3 (see Irving & McGlynn 1976; Cavenaugh & Seyfert 1977) is therefore biased towards the exclusion of pre-1850 Ma rock units which may possess post-1850 Ma magnetizations if this point is not adequately clarified.

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Table 3. Palaeomagnetic poles from the structural/age provinces of the Laurentian Shield for the interval *c.* 1850–1600 Ma.

Pole No.	Rock Unit	Age	Palaeomagnetic Pole	Reference **
(i)	Labrador and Nain Province			
1.	Murdoch, Thompson Lake and Menihek metavolcanics	1816*	2 S, 197 E	
2.	Seward Subgroup	1800***	(i) 20 S, 265 E (ii)26 S, 267 E	Park (1977) Park (1977)
3.	Menihek 'B' magnetisations	"	6 N, 279 E	" "
4.	Seward 'B' magnetisations	"	5 N, 251 E	" "
5.	Retty 'B' magnetisations	"	(i) 28 S, 290 E (ii)31 S, 278 E	" " " "
(ii)	NW Scotland and Greenland			
	L1 Lewisian 'A' magnetisations	1650 - 1625	38 N, 235 E	This paper
	L2 Lewisian 'B' magnetisations		(i) 5 S, 245 E (ii)2 S, 195 E	" " " "
	L3 Lewisian 'C' magnetisations		42 N, 78 E	" "
6.	Nordre Stromfjord area	——	27 N, 263 E	Beckmann et al. (1977)
7.	Holsteinborg area	——	27 N, 286 E	" "
8.	Itivdleq area dykes	1830 - 1712+	(i) 16 N, 271 E	Beckmann & Mitchell(1976)
		" "	(ii)13 N, 269 E	Morgan (1976 a and b)
9.	Sagdlerssuaq	1620+	57 N, 266 E	Beckmann & Mitchell(1976)
10.	Ketilidian metavolcanics	1730 - 1600+	(i) 58 N, 271 E	Piper & Stearn (1976)
		" "	(ii)64 N, 221 E	" "
11.	Kangamiut dykes	1790 - 1620+	22 N, 251 E	Fahrig & Bridgwater(1976)
12.	Amphibolite boudins	——	57 N, 220 E	" "
13.	Gȣdhab gneisses	——	25 N, 250 E	" "
(iii)	Superior Province			
14.	Nipissing diabase, 'E' magnetisations	1800***	(i) 9 N, 1 E (ii)8 S, 23 E (iii)1 N, 31 E	Roy & Lapointe (1976) Patel and Palmer (1974) Symons (1971)
(iv)	Beartooth Uplift			
15.	Wind River dykes	1880 - 1680+	43 N, 239 E	
(v)	Churchill and Bear Structural Provinces			
16.	Stark Formation	1845 - 1630+	15 S, 215 E	
17.	Tochatwi "	1845 - 1818+	18 S, 216 E	
18.	Metamorphosed Kaminak dykes	1892 - 1615+	20 N, 268 E	
19.	Flin-Flon, secondary magnetisations	1800***	20 S, 314 E	
20.	Flin Flon, tertiary magnetisations	1700 - 1600+	24 N, 265 E	
21.	Et-Then Group	1835 - 1630+	1 S, 312 E	
22.	Martin Formation	1930 - 1635+,++	9 S, 288 E	

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Table 3 – *continued*

23.	Kahocella, primary magnetisation	1873^{++}	6 N, 288 E		
24.	Kahocella, secondary magnetisation	<1873	18 N, 266 E		
25.	Dubawnt Group	1787*	7 N, 277 E		
26.	Nonacho sediments	>1700	13 N, 274 E		
27.	Sparrow dykes	1700^{+}	12 N, 291 E		
28.	Daly Bay 'A' magnetisations	1622^{+}	12 N, 259 E		
29.	Cape Smith metavolcanics	1650 – 1460^{+}	16 N, 253 E		
30.	Pearson Formation 'A' magnetisations	1800***	19 S, 283 E	McGlynn & Irving (1978)	
31.	" " 'B' magnetisations	——	5 S, 326 E	"	"
32.	" " 'C' magnetisations	——	6 N, 293 E	"	"
33.	" " 'D' magnetisations	——	27 N, 245 E	"	"
(vi)	Slave Province				
34.	Western Channel diabase	1700 – 1400^{+}	9 N, 245 E		

Footnotes

All specific ages have been recalculated to the now internationally agreed decay constants listed by Steiger & Jäger (1977). ** Poles not referenced here are listed and referenced by Irving & Hastie (1975) and Cavenaugh & Seyfert (1977). *** Magnetizations considered to have been acquired during uplift following the Hudsonian mobile episode (*c.* 1800 Ma). + = K–Ar mineral, whole rock or isochron ages, * = Rb–Sr whole rock, mineral or isochron ages; for further details and summary of age data, the reader is referred to Cavenaugh & Seyfert (1977). K–Ar ages are quoted to the nearest 10 Ma. The Greenland poles are listed after rotation of 22° clockwise about a Euler pole at 73° N, 98° E, and the NW Scotland poles are listed after rotation of 38° about a Euler pole at 88.4° N, 27.7° E.

A most important study relating to Lower Proterozoic times is that of Roy & Lapointe (1976) who identify five magnetizations (A → E) from the Superior Province of which the last three are present in the Nipissing diabase dated 2117 ± 27 and 2110 ± 80 Ma (Rb-Sr isochron ages); the E magnetization is remote from the earlier magnetizations and is believed to have been acquired during the Hudsonian mobile episode (*c.* 1800 Ma). Poles 1, 16 and 17 (Table 3, Fig. 8) from other Provinces correspond closely with the E magnetizations; the mutual agreement between these poles renders a tectonic explanation for poles 16 and 17 (Evans & Bingham 1976; Bingham & Evans 1976) less likely. Although the age data relating to this group is not definitive, these data suggest that it is *c.* 1850–1800 and therefore slightly older than the remaining poles in Fig. 8. These latter poles plot in a quite different field and are in part derived from the same provinces; this large difference must leave some doubt about the polarity of the pre-1800 Ma part of the apw curve (see Fig. 8 and Roy & Lapointe 1976).

The 1800 to 1650 Ma poles define a SE to NW trending swathe. The direction of polar movement along this swathe is now well defined by the migration of high to low blocking temperature components in dykes and gneisses of west Greenland (Morgan 1976a, b), the Flin Flon primary and secondary magnetizations (poles 19 and 20, Fig. 8), the Pearson Formation A, B and C magnetizations (30–32, Fig. 8) and possibly by the Kahocella primary and secondary magnetizations (23 and 24, Fig. 8) for the Churchill and Bear structural provinces. There is also good agreement between the age estimates of the poles

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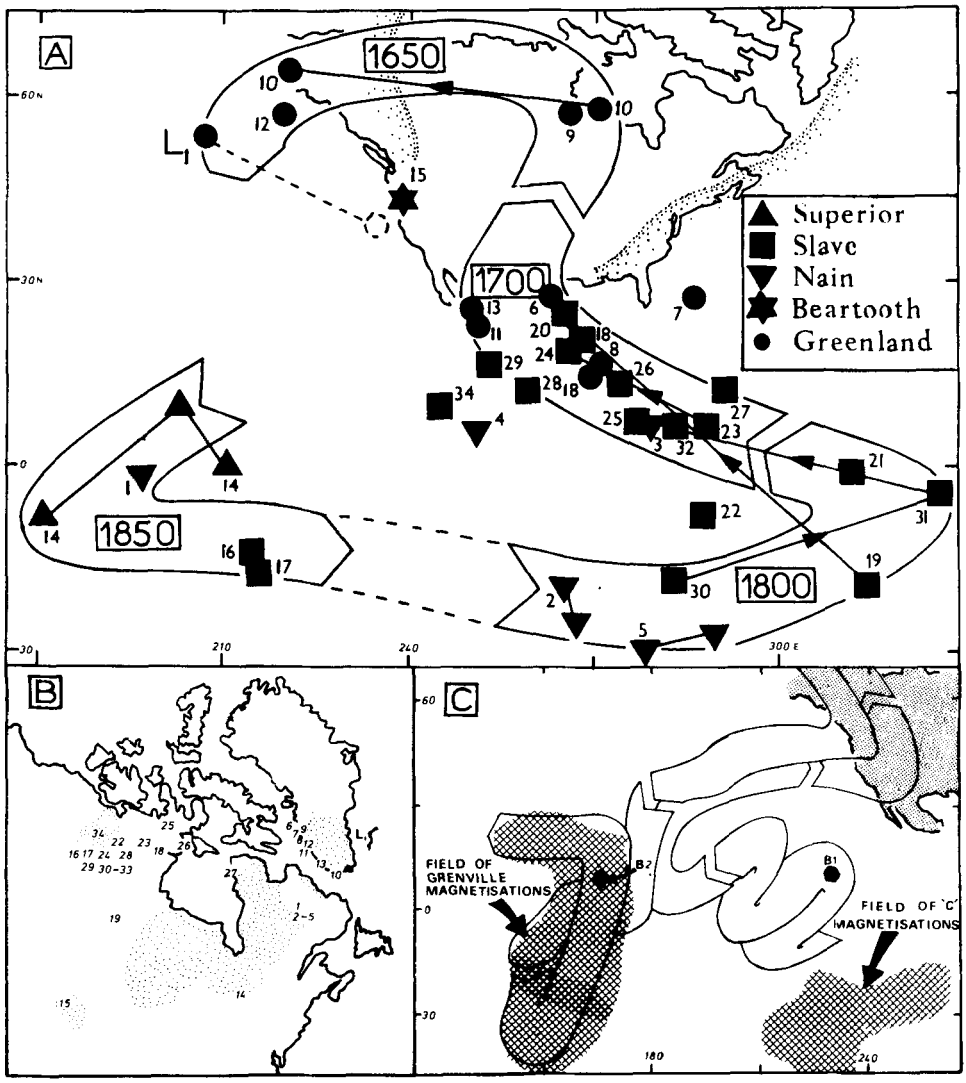


Figure 8. (a) Apparent polar wander curve for the Laurentian Shield derived from the data for the constituent structural/age provinces listed in Table 3; the ages given on this curve are only general indicators and the reader is referred to Table 3 for the primary data; poles from the same rock suite are connected and arrows indicate the age relationship where this is known. The palaeomagnetic pole derived from the Lewisian 'A' magnetizations (L1) is plotted before and after a 30° correction for local rotation. (b) The location of the studies yielding the palaeomagnetic poles of Table 3 and (a) on the reconstruction of the Laurentian craton; the Archaean nuclei are stippled. (c) This figure examines the possible correlation of the 'B' and 'C' magnetizations with the Upper Proterozoic apw path for the Laurentian Shield after Irving *et al.* (1977) and Piper (1977); the field of Grenville magnetizations (*c.* 1000–700 Ma) is taken from McWilliams & Dunlop (1978).

from the Churchill/Bear provinces and west Greenland: rocks from the former regions acquired their magnetizations during uplift and cooling following the Hudsonian mobile episode and are linked to a range of K–Ar and Rb–Sr ages mostly lying within the range 1800–1700 Ma. Some of the poles from Greenland were originally regarded as late Archaean or early Proterozoic in age (Fahrig & Bridgwater 1976) but a Rb–Sr isochron of 1909 ±

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60 Ma (Kalsbeek, Bridgwater & Zeck 1978) on the Kangamiut dykes defines a maximum age for the magnetizations because, like the Scourie dykes, the west Greenland dyke swarms were intruded into hot crust (Escher *et al.* 1976) and their magnetic blocking temperature ranges were greatly depressed during slow uplift and cooling (Morgan 1976a, b); these poles are now linked to K–Ar mineral and isochron ages between 1800 and 1660 Ma. The youngest part of this swathe is defined by magnetizations from the Ketilidian mobile belt of south Greenland linked to K–Ar and Rb–Sr mineral ages of 1700–1600 Ma (Van Breemen, Aftalian & Allaart 1974; Piper & Stearn 1976) which also agree with two other poles from west Greenland (9 and 12) of which the Sagdlerssuaq amphibolites have a K–Ar isochron age of 1620 ± 50 Ma. This SE–NW trending part of the apw swathe is thus confined between the limits *c.* 1800–1600 Ma and the pole for the Lewisian 'A' magnetizations lies at the younger end of this swathe.

Several general conclusions can be drawn from the Middle Proterozoic palaeomagnetic data of Table 3 and Fig. 8(a):

(1) Although Cavenaugh & Seyfert (1977) have utilized pre-1600 Ma data from the Laurentian Shield to define separate apw curves for the Slave and Superior cratons, and have proposed that the Slave and part of the Churchill province joined the remainder of the Shield by plate collision in the Hudsonian mobile episode, their case has been strongly criticized by Roy *et al.* (1978). In addition to the arguments advanced by these latter authors, the hypothesis of Cavenaugh & Seyfert is not tenable because the poles they utilise to define an apw path for the Slave craton are coincident with the large body of contemporaneous Greenlandic data (Fig. 8) from the opposite side of the Hudsonian mobile zone.

(2) These magnetizations from the northern Churchill/Slave region are in any case not relevant to the tectonic origin of the Hudsonian mobile episode because at least the majority post-date the tectonic activity associated with this episode. As with the later Grenville episode (*c.* 1100 Ma), the palaeomagnetic data from the mobile zone itself cannot be used to refute an origin by plate convergence because they entirely post-date the mobile belt tectonism. However, the agreement of poles from the Superior (14), Churchill (16, 17) and Nain (1) provinces (Fig. 8) and certain older data (Irving & McGlynn 1976; Roy *et al.* 1978) are strong evidence that the Laurentian Shield was already an integral unit by 1850 Ma.

(3) Although there is good overall agreement between these mid-Proterozoic palaeomagnetic data from the constituent provinces of the Laurentian craton, there are some critical differences and it emerges that different parts of the Shield provide complementary information to piece together the overall apw record. Thus the earliest part of the SE–NW swathe is well defined by a range of primary and secondary magnetizations from the Churchill and Bear provinces; these overlap with the oldest poles from Greenland which in turn extend the apw curve into a zone not recognized in the Churchill/Bear provinces. The youngest poles derive from the Ketilidian and Lewisian rocks from the southern margin of the Laurentian Shield which were being uplifted and cooled at *c.* 1650 Ma; comparable magnetizations to these have not yet (with the possible exception of pole 15) been recognized in North America.

(4) No current data from the Laurentian Shield are known with certainty to relate to the interval 1650–1450 Ma; immediately after this the apw curve is defined by a combination of North American (Irving & McGlynn 1976; Irving, Emslie & Park 1977) and Greenlandic (Piper 1977) results from Upper Proterozoic intrusive and supracrustal suites. The Lewisian 'B' magnetizations of mixed polarity correlate with these post-1450 Ma results (Fig. 8(c)).

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The Lewisian 'C' magnetizations present a greater problem: the palaeomagnetic pole both before (258° E, 42° S) and after (229.4° E, 27.6° S) local rotation of the Lewisian complex is accommodated does not correlate with any relevant Laurentian data (Fig. 8(c)). It does not correlate with magnetizations from the Grenville Province (generalized in Fig. 8(c) after McWilliams & Dunlop 1978) although a thermal/tectonic episode of this age is recognized in the Moinian rocks to the east of the Moine Thrust zone (e.g. Brook, Brewer & Powell 1976). It is probable that these magnetizations record later stages of the regional uplift when the low blocking temperature phases were acquiring their remanence, and they relate to the interval 1650–1450 Ma for which the apw curve is, as yet, undefined.

(5) As noted by Park (1975) metamorphic palaeomagnetism presents a range of problems to which solutions may only be expected to emerge gradually. In particular, it is not yet completely clear why some magnetizations (a) are well represented while others are poorly defined (b) or of complex origin (c). It seems likely that such factors as initial compositions, variable cooling rates and proximity to shear zones may be important to this problem but they must await detailed studies for clarification.

(6) The analysis of Table 3 and Fig. 8 has been careful to exclude from consideration poles which are older than 1850 Ma. Earlier analyses (e.g. Irving & Park 1972b) have attempted to compress all palaeomagnetic poles assigned to the interval 2200–1700 Ma into a single closed apw loop. The discovery of Nipissing diabase 'E' magnetizations by Roy & Lapointe (1976) showed that this was an oversimplified approach and resolution of this point is critical to correlations between the Precambrian shields (Piper 1976). In the context of the Laurentian data, it is important to distinguish those pre-1850 Ma results mostly from the large Superior craton where it is known from the overlying pre-2300 Ma Huronian sediments that present erosion levels had already been reached, and those post-1850 Ma results from most of the remainder of the shield. These latter results include the Lewisian, Greenlandic and Churchill data; although these rocks are often of Archaean origin, the present erosion levels were at such great crustal depths during the widespread *c.* 1800 Ma mobile activity that there is no possibility of an older palaeomagnetic record within them.

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